

10 The golden poison dart frog, *Phyllobates terribilis*, lives in the Colombian rainforest ecosystem.

Fig. 10.1 shows golden poison dart frogs.



Fig. 10.1

(a) Define the term ecosystem.

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(b) One way of estimating the size of a population of golden poison dart frogs is to use the mark-release-recapture method.

(i) Suggest the assumptions that must be made for the mark-release-recapture method to be valid.

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(ii) A first sample of 27 golden poison dart frogs was captured, marked and released. When a second sample of 33 frogs was captured, 13 had marks on them.

Use the Lincoln index to estimate the population size of the frogs.

$$\text{estimate of population size} = \frac{n_1 \times n_2}{m_2}$$

n_1 = number of individuals captured in first sample

n_2 = number of individuals (both marked and unmarked) captured in second sample

m_2 = number of marked individuals recaptured in second sample

answer [2]

[Total: 8]